

Object and classes

A. Tick (✓) the correct option.

1. Which among the following creates a blueprint to represent characteristic and behaviour of an object?
- | | |
|-------------|------------------|
| a. Object | b. Class |
| c. Instance | d. None of these |

Ans. b. Class

2. Which among the following do not belong to an object?
- | | |
|-------------|--------------|
| a. State | b. Behaviour |
| c. Identity | d. Class |

Ans. a. State

3. Which among the following is not a component of a class?
- | | |
|--------------|---------------|
| a. Modifiers | b. Class name |
| c. Object | d. Body |

Ans. c. Object

4. Which among the following keyword is used to allocate memory space for an object?
- | | |
|----------|--------|
| a. new | b. for |
| c. while | d. int |

Ans. a. new

5. Which among the following operator is used to access individual members of an object?
- | | |
|--------------|---------------|
| a. . (dot) | b. + (plus) |
| c. – (minus) | d. / (divide) |

6. Which among the following modifier is used in a 'class'?
- | | |
|-----------------|------------------|
| a. public | b. default |
| c. Both a and b | d. None of these |

Ans. c. Both a and b

7. Which among the following is a valid class name?

- a. Simple Interest
- b. SimpleInterest
- c. 1SimpleInterest
- d. Simple@Interest

Ans. b. SimpleInterest

8. Which among the following is a valid object name?

- a. obj1
- b. 1obj
- c. Obj 1
- d. Obj#1

Ans. a. obj1

9. Which among the following is used to represent behaviour in a class?

- a. Method
- b. Data members
- c. Both a and b
- d. None of these

Ans. a. Method

10. If a method named show() is to be invoked using an object 'ob' , which among the following is correct?

- a. ob.show()
- b. ob.show
- c. show().ob
- d. None of these

Ans. a. ob.show()

B. State whether the following statements are True (T) or False (F).

- 1. An object is called a class factory.
- 2. A class is an instance of an object.
- 3. A class is a mechanism to implement encapsulation.
- 4. Data members in a class is used to represent the characteristic of an object.
- 5. The new operator is used to create an object.
- 6. It's a rule to have a class-name beginning in capital letter.
- 7. Java is case sensitive.
- 8. A class-name cannot be a keyword.
- 9. The dot operator is used to access members in an object using an object.
- 10. In programming every object will have a name.

- F
- F
- T
- T
- T
- F
- T
- T
- T
- T

C. Fill in the blanks.

- 1. A class is a template that binds together data and methods together.
- 2. The values in the attributes of an object is called the state of an object.
- 3. The .dot operator is used to access the individual members of a class.

4. The keyword new is used to allocate memory space for an object.
5. The default and public access modifier is used with a class.
6. It is a common convention to begin a class-name in capital letter.
7. A class is called an object factory.
8. Object-name is used to give a unique name to an object for identification.
9. Behaviour of an object is represented by methods.
10. The size of fundamental data type is fixed.

1. State two differences between fundamental and user-defined data type.

Ans.

Comparison of Fundamental & User defined data type.	
Fundamental data type	User defined data type
These are inbuilt data type provided by the Java Language.	These are data types created by the user using fundamental or user defined data type or both.
The size of it is fixed.	The size of different user-defined data type depends upon the size of the individual components of it.
These data types are available in all parts of a program within a class.	These data types are available only as specified by the access specifiers.

2. State two differences between a class and an object.

Ans.

A class	An object
Class is a blueprint or template from which objects are created.	Object is an instance of a class.
Class is a group of similar objects .	Object is a real world entity such as pen, laptop, mobile, bed, keyboard, mouse, chair etc.
Class is a logical entity.	Object is a physical entity.

3. Why is a class called a user-defined data type?

Ans. Primitive **data types** are the general and fundamental **data types** that we have in **Java** and those are byte, short, int, long, float, double, char, boolean, etc., **User defined data types** are those that **user / programmer** himself defines.

4. What is an access specifier? Which two access specifier is used in a class declaration?

Ans. **Access modifiers (or access specifiers)** are keywords in object-oriented languages that set the accessibility of classes, methods, and other members. **Access modifiers** are a specific part of programming language syntax used to facilitate the encapsulation of components.

The public and default access specifier is used in a class declaration.

8. State two conventions you should follow while naming a class.

Ans. Class name should begin with capital letter. Example, Bank, School, Student

If a class name consists of more than one word, the first letter of each new word should be in uppercase. For example MyClass, AccountDetails, SimpleInterest, etc.

9. Write a statement in Java that will declare an object named si of the SimpleInterest class.

Ans. SimpleInterest si=new SimpleInterest();

11. Which among the following are invalid class name in Java? State with reasons.

1. Compound Interest

Ans. Invalid as it contains a space.

2. 1MyClass

Ans. Invalid as it should not begin with a digit.

3. MyClass\$

Ans. Valid

4. Myclass#

Ans. Invalid as it contains a special character #.

5. My@Class

Ans. Invalid as it contains a special character @.

Keywords to focus on: Instance, Instantiation, object, object factory, class, blueprint access specifiers (default, public, private, protected) , Syntax to create object

public: Most accessibility

private: least accessibility

protected: accessible by child class or sub class or derived class

default: behaves like public, used when no access specifier is written